

Easy (10)

1.

What is the default violation action in port security?

- A) Shutdown
- B) Restrict
- C) Protect
- D) Alert

Answer: A) Shutdown

Explanation: By default, if a violation occurs on a port security-enabled interface, the port is put into an error-disabled (shutdown) state to block unauthorized devices.

2.

Which command enables port security on an interface?

- A) switchport port-security
- B) switchport access vlan
- C) ip access-group
- D) enable port-security

Answer: A) switchport port-security

Explanation: This command enables port security features on the interface.

3.

What is the purpose of DHCP snooping?

- A) Prevent unauthorized DHCP servers
- B) Encrypt DHCP traffic
- C) Limit number of DHCP clients
- D) Block ARP requests

Answer: A) Prevent unauthorized DHCP servers

Explanation: DHCP snooping blocks rogue DHCP servers by allowing only trusted DHCP servers to respond to DHCP requests.

4.

Which protocol encrypts the entire packet between device and AAA server?

- A) RADIUS
- B) TACACS+
- C) PAP
- D) CHAP

Answer: B) TACACS+

Explanation: TACACS+ encrypts the entire payload, unlike RADIUS which encrypts only the password field.

5.

What does ACL stand for?

- A) Access Control List
- B) Automatic Control List
- C) Access Connection List
- D) Advanced Control Layer

Answer: A) Access Control List

Explanation: ACLs are lists of rules used to permit or deny traffic based on various parameters.

6.

Which ACL type filters traffic based on Layer 3 and Layer 4 info?

- A) Standard ACL
- B) Extended ACL
- C) VLAN ACL
- D) MAC ACL

Answer: B) Extended ACL

Explanation: Extended ACLs filter traffic based on source/destination IP, protocol, and ports, which are Layer 3 and 4 details.

7.

Which command applies an IPv6 ACL inbound on an interface?

- A) ip access-group
- B) ipv6 access-list
- C) ipv6 traffic-filter
- D) ipv6 access-group

Answer: C) ipv6 traffic-filter

Explanation: The command `ipv6 traffic-filter [ACL]` in applies an IPv6 ACL inbound on an interface.

8.

What command disables HTTP server on a Cisco device?

- A) no ip http server
- B) no service http
- C) disable http
- D) shutdown http

Answer: A) no ip http server

Explanation: This command disables the insecure HTTP server on the device for hardening.

9.

Which command is used to enable SSH access?

- A) transport input ssh
- B) enable ssh
- C) ip ssh enable
- D) ssh enable

Answer: A) transport input ssh

Explanation: This command on VTY lines restricts remote access to only SSH (more secure than Telnet).

10.

What is the default VTP mode on Cisco switches?

- A) Server
- B) Client

C) Transparent

D) Off

Answer: A) Server

Explanation: By default, Cisco switches run in VTP server mode, allowing VLAN configuration changes to propagate.

Medium (30)

11.

Which port security violation mode drops packets without logging?

A) Shutdown

B) Restrict

C) Protect

D) Block

Answer: C) Protect

Explanation: Protect mode silently drops packets from violating MAC addresses but does not generate logs or shut down the port.

12.

Which command shows the port security status on an interface?

A) show port-security interface [interface]

B) show running-config

C) show vlan

D) show mac address-table

Answer: A) show port-security interface [interface]

Explanation: This command shows detailed port security info including violation counts, secure MAC addresses, and status.

13.

Which access layer mitigation blocks rogue DHCP servers?

- A) DHCP Snooping
- B) Dynamic ARP Inspection
- C) IP Source Guard
- D) Storm Control

Answer: A) DHCP Snooping

Explanation: DHCP Snooping filters DHCP traffic and blocks untrusted DHCP servers.

14.

What does IP Source Guard use to filter traffic?

- A) MAC address only
- B) IP and MAC address binding
- C) VLAN ID only
- D) Port number only

Answer: B) IP and MAC address binding

Explanation: IP Source Guard checks DHCP snooping binding table to verify IP-MAC-port mappings to block spoofing.

15.

Which AAA component verifies user identity?

- A) Authentication
- B) Authorization
- C) Accounting
- D) Access

Answer: A) Authentication

Explanation: Authentication confirms the identity of a user or device.

16.

Which AAA component controls user permissions?

- A) Authentication
- B) Authorization
- C) Accounting
- D) Access

Answer: B) Authorization

Explanation: Authorization defines what resources or commands a user is allowed to access.

17.

What UDP port does RADIUS use?

- A) 1812
- B) 22

- C) 49
- D) 23

Answer: A) 1812

Explanation: RADIUS uses UDP 1812 for authentication and UDP 1813 for accounting.

18.

What TCP port does TACACS+ use?

- A) 49
- B) 1812
- C) 22
- D) 23

Answer: A) 49

Explanation: TACACS+ uses TCP port 49.

19.

Which command is used to configure a static MAC address for port security?

- A) switchport port-security mac-address [MAC]
- B) mac-address-table static
- C) ip mac-address
- D) static mac-address

Answer: A) switchport port-security mac-address [MAC]

Explanation: This command sets a specific MAC address allowed on a secure port.

20.

What is the maximum number of MAC addresses on a secure port if not configured?

- A) 1
- B) 2
- C) Unlimited
- D) 0

Answer: A) 1

Explanation: By default, port security allows only 1 MAC address.

21.

Which ACL matches traffic only by source IP?

- A) Standard ACL
- B) Extended ACL
- C) Named ACL
- D) MAC ACL

Answer: A) Standard ACL

Explanation: Standard ACLs filter traffic based only on source IP addresses.

22.

What is the first step when troubleshooting an ACL?

- A) Check interface ACL application
- B) Remove ACL
- C) Reboot device
- D) Enable logging

Answer: A) Check interface ACL application

Explanation: Verify that the ACL is correctly applied to the right interface and direction.

23.

What does the APIC-EM Path Trace tool analyze?

- A) Network path and ACL effects
- B) CPU usage
- C) Switch configuration
- D) Routing table

Answer: A) Network path and ACL effects

Explanation: It helps visualize the path of packets and how ACLs impact traffic flow.

24.

Which feature prevents VLAN hopping attacks?

- A) Change native VLAN from 1
- B) Enable DTP
- C) Enable VTP
- D) Disable STP

Answer: A) Change native VLAN from 1

Explanation: VLAN hopping exploits the default native VLAN (1); changing it prevents such attacks.

25.

What does the switchport nonegotiate command do?

- A) Disables DTP negotiation
- B) Enables negotiation
- C) Sets port to trunk
- D) Sets port to access

Answer: A) Disables DTP negotiation

Explanation: Prevents the switch from sending DTP frames and negotiating trunks.

26.

Which port security violation mode generates log and does not shutdown port?

- A) Restrict
- B) Shutdown
- C) Protect
- D) Block

Answer: A) Restrict

Explanation: Restrict mode drops violating frames and logs violations without shutting down port.

27.

What type of ACL filters based on source/destination IP, port, protocol?

- A) Extended ACL
- B) Standard ACL
- C) MAC ACL
- D) VLAN ACL

Answer: A) Extended ACL

Explanation: Extended ACLs filter on Layer 3 and Layer 4 parameters.

28.

How do you verify if a TACACS+ server is reachable?

- A) ping the server IP
- B) show tacacs
- C) Both A and B
- D) show radius

Answer: C) Both A and B

Explanation: Ping tests connectivity; show tacacs shows TACACS+ status.

29.

What is the effect of enabling BPDU Guard on a port?

- A) Shuts down port receiving BPDU
- B) Enables STP
- C) Blocks DHCP
- D) Enables VLAN tagging

Answer: A) Shuts down port receiving BPDU

Explanation: BPDU Guard disables ports that receive unexpected BPDU frames to protect STP topology.

30.

What command sets the enable secret password?

- A) enable secret [password]
- B) password enable [password]
- C) set enable [password]
- D) secret enable [password]

Answer: A) enable secret [password]

Explanation: This command sets an encrypted enable password.

Hard (10)

31.

What happens if a violation occurs on a port configured with protect mode?

- A) Frames from violating MAC addresses are dropped silently
- B) Port shuts down
- C) Violation is logged and packets dropped
- D) Port reboots

Answer: A) Frames from violating MAC addresses are dropped silently

Explanation: Protect mode silently drops violating frames without logs or shutdown.

32.

What protocol field does TACACS+ encrypt?

- A) Entire packet
- B) Only password field
- C) Only header
- D) None

Answer: A) Entire packet

Explanation: TACACS+ encrypts the entire payload, enhancing security.

33.

What is the difference between standard and extended ACLs regarding placement?

- A) Standard ACLs near destination, Extended near source
- B) Standard ACLs near source, Extended near destination
- C) Both anywhere
- D) Standard near source only

Answer: A) Standard ACLs near destination, Extended near source

Explanation: Standard ACLs filter only by source IP, so placed near destination; extended ACLs filter on source/dest and ports, placed near source.

34.

Which command verifies active sessions using TACACS+?

- A) show aaa sessions
- B) show tacacs
- C) show users
- D) show sessions

Answer: A) show aaa sessions

Explanation: Displays current AAA sessions including TACACS+.

35.

What is the purpose of login local command on VTY lines?

- A) Uses local user database for authentication
- B) Enables remote login without password
- C) Disables login
- D) Enables TACACS only

Answer: A) Uses local user database for authentication

Explanation: This command tells the device to use locally configured users for remote login.

36.

Which IPv6 ACL command applies filter inbound on an interface?

- A) ipv6 traffic-filter [acl] in
 - B) ipv6 access-group [acl] in
 - C) ip access-group [acl] in
 - D) ipv6 access-list apply
- Answer:** A) ipv6 traffic-filter [acl] in

Explanation: This command applies an IPv6 ACL on incoming traffic.

37.

How does IP Source Guard determine allowed traffic?

- A) DHCP snooping binding database
- B) MAC address table only
- C) VLAN database
- D) Static ACLs only

Answer: A) DHCP snooping binding database

Explanation: IP Source Guard uses DHCP snooping bindings to verify legitimate IP-MAC-port pairs.

38.

What is the default login authentication method if AAA is not configured?

- A) None
- B) Local
- C) TACACS+
- D) RADIUS

Answer: A) None

Explanation: Without AAA, login authentication defaults to none (no authentication).

39.

Which feature can protect a switch from MAC flooding attacks?

- A) Port security
- B) DHCP snooping
- C) DAI (Dynamic ARP Inspection)
- D) IP Source Guard

Answer: A) Port security

Explanation: Port security limits the number of MAC addresses on a port, preventing MAC flooding.

40.

What happens if you apply an ACL with no permit statements?

- A) All traffic is denied
- B) All traffic is allowed
- C) Device crashes
- D) ACL is ignored

Answer: A) All traffic is denied

Explanation: ACLs have an implicit deny at the end; no permit means deny all.

41.

Which command disables unused services to harden device?

- A) no ip http server
- B) disable services

C) no service disable

D) shutdown services

Answer: A) no ip http server

Explanation: Disabling unused services like HTTP server reduces attack surface.

42.

What AAA method provides accounting of user actions?

A) Accounting

B) Authentication

C) Authorization

D) Access

Answer: A) Accounting

Explanation: AAA accounting logs user actions for auditing.

43.

Which protocol uses UDP port 1813 for accounting?

A) RADIUS

B) TACACS+

C) PAP

D) CHAP

Answer: A) RADIUS

Explanation: RADIUS uses UDP 1813 for accounting messages.

44.

How many MAC addresses can be configured with sticky learning on a port?

- A) Up to the maximum configured
- B) Only 1
- C) Unlimited
- D) None

Answer: A) Up to the maximum configured

Explanation: Sticky learning dynamically learns and saves MACs up to the max allowed.

45.

Which command is used to enable port security aging?

- A) switchport port-security aging time [minutes]
- B) port-security aging enable
- C) enable port-security aging
- D) port-security aging time

Answer: A) switchport port-security aging time [minutes]

Explanation: This command configures aging time for learned MAC addresses.

46.

What is the purpose of TACACS+ Authorization?

- A) Controls what commands a user can run
- B) Authenticates user
- C) Logs user activity
- D) Encrypts password

Answer: A) Controls what commands a user can run

Explanation: Authorization defines permissions and command privileges.

47.

What is the result of a misconfigured ACL on a router interface?

- A) Legitimate traffic might be blocked
- B) Traffic flows unrestricted
- C) Router crashes
- D) Interface shuts down

Answer: A) Legitimate traffic might be blocked

Explanation: ACL errors can unintentionally block valid traffic causing outages.

48.

Which Cisco feature monitors real-time network activity and can be secured with AAA?

- A) Cisco IOS CLI
- B) SNMPv1
- C) FTP
- D) HTTP

Answer: A) Cisco IOS CLI

Explanation: The CLI is secured via AAA to control device access.

49.

What is the function of BPDU Guard?

- A) Protects STP topology by disabling ports receiving BPDUs unexpectedly
- B) Prevents DHCP attacks
- C) Filters MAC addresses
- D) Controls VLAN traffic

Answer: A) Protects STP topology by disabling ports receiving BPDUs unexpectedly

Explanation: BPDU Guard disables ports to prevent potential STP loops.

50.

What does the command `aaa authorization exec default group tacacs+ local` do?

- A) Authorizes user exec sessions via TACACS+, fallback to local
- B) Authenticates users locally only
- C) Authorizes users without authentication
- D) Disables authorization

Answer: A) Authorizes user exec sessions via TACACS+, fallback to local

Explanation: This config attempts TACACS+ authorization first, then falls back to local.